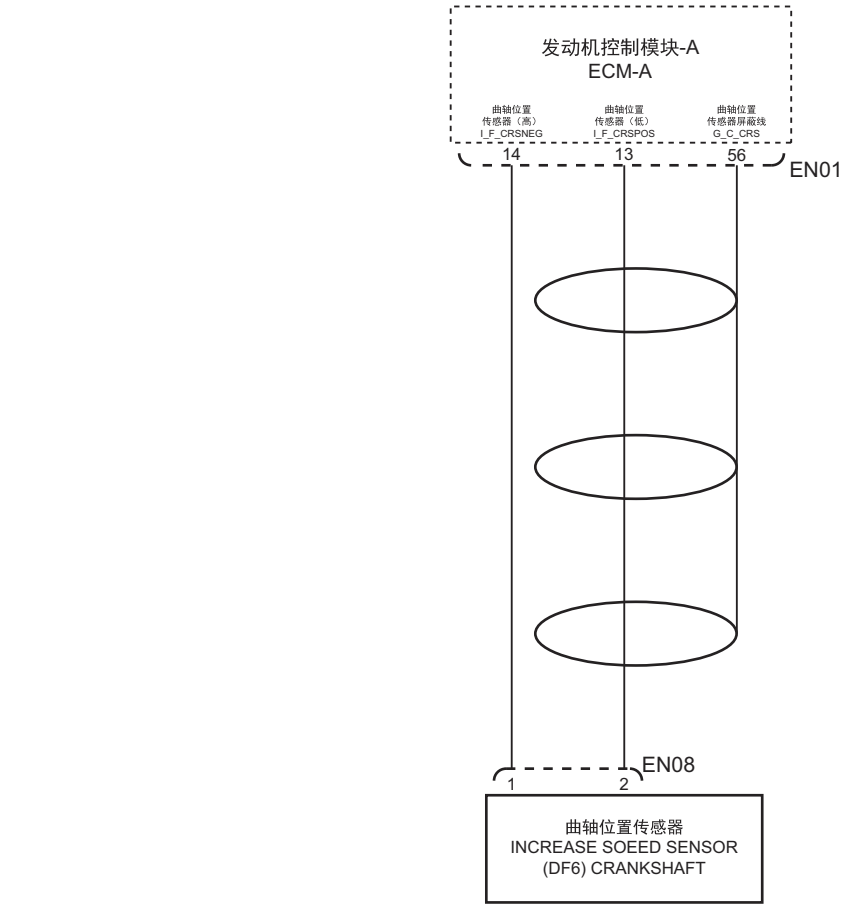
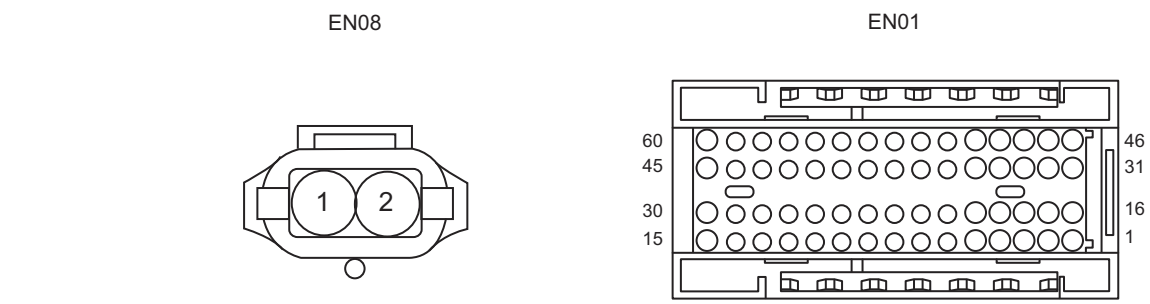
P0335、 P0336、 P0219

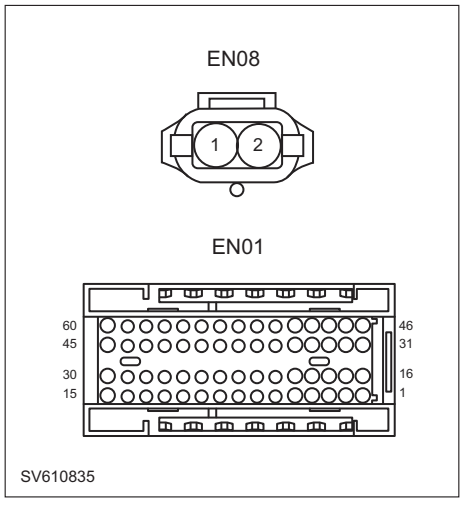
Description of fault diagnosis codes
P0335-31：DFC for crankshaft signal diagnose - no signal
P0336-29：DFC for crankshaft signal diagnose - disturbed signal
P0219-00：Overspeed detection in component engine protection
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Harness or plug (open circuit or short circuit on crankshaft position sensor circuit) • Signal panel error • Gap between crankshaft position sensor and signal panel • Crankshaft position sensor
Description of circuit diagram
The crankshaft sensor is used to calculate the engine speed and the crankshaft position. Such sensor is also applicable to other speed detection system (e.g. gear and engine speed sensor).

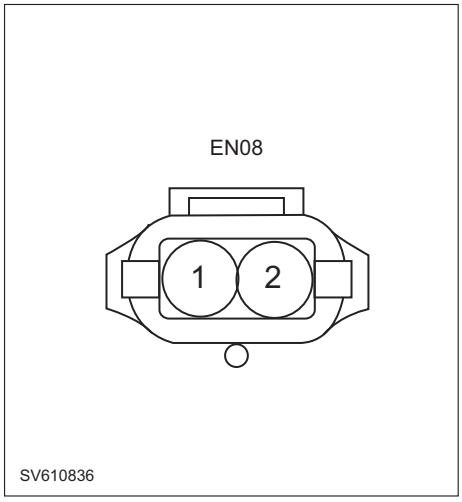
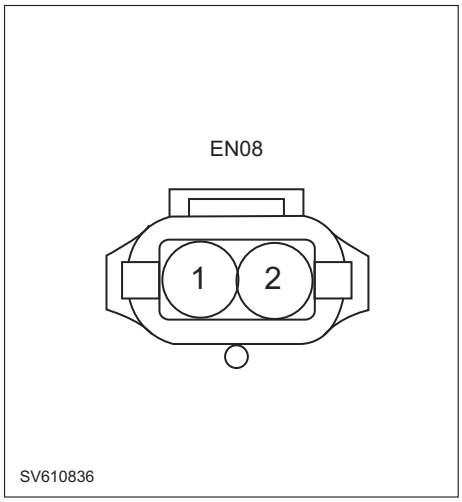
Circuit diagram





SV610834

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check if the gap between crankshaft sensor and signal panel is normal. 4 Check the signal panel. 5 Check if there is any damage, bad contact, aging and looseness phenomenon on harness plug of engine control module and crankshaft position sensor. 6 Check whether the result is normal?
		No → Repair or replace the faulted parts.
Yes ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON. 3 Diagnose the engine control system with diagnostic apparatus. 4 If there is any fault code except for P0335/P0336/P0219?
		Yes → Refer to: DTC Summary List (EU5).
No ↓		
Step 3	Check if the harness plugs of crankshaft position sensor and engine control module are in open circuit.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of crankshaft position sensor. 3 Disconnect the harness plug of engine control module. 4 Measure the resistance between harness plugs of crankshaft position sensor and engine control module. Measuring circuit: resistance between terminal 1 of plug EN08 and terminal 14 of plug EN01. Measuring circuit: resistance between terminal 2 of plug EN08 and terminal 13 of plug EN01. **Standard value: ≈ 0Ω** 5 Check whether the result is normal?
		No → Repair or replace the faulted parts.
Yes ↓		

Step 4	Check the circuit (short circuit) of crankshaft position sensor.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of crankshaft position sensor. 3 Disconnect the harness plug of engine control module. 4 Measure the resistance between the harness plug terminal of crankshaft position sensor and ground. Measuring circuit: the resistance between terminal 1 of plug EN08 and ground. Measuring circuit: the resistance between terminal 2 of plug EN08 and ground. **Standard value: ∞** 5 Measure the voltage between the harness plug terminal of crankshaft position sensor and ground. Measuring circuit: the voltage between terminal 1 of plug EN08 and ground. Measuring circuit: the voltage between terminal 2 of plug EN08 and ground. **Standard value: ≈ 0V** 6 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 5	Check crankshaft position sensor (component inspection).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of crankshaft position sensor. 3 Measure the resistance between the terminals of crankshaft position sensor. Measuring circuit: resistance between terminal 1 and terminal 2 of plug EN08. **Standard value: 770~950Ω (at 20°C)** 4 Check whether the result is normal?
	No → Replace the crankshaft position sensor.
Next ↓	
Step 6	Complete the check and confirm whether the system works normally nor not.